

The Silent Tax: Typo-Robustness Analysis (GSM8K_20000)

DeepSeek-R1-Distill-Qwen-7B

All tables are *answered-only*: traces that hit the token cap before writing a final answer are excluded from the metrics. Configs are named `typo{rate}_real{ratio}`: *rate* = % of words corrupted, *real* = % of typos that form real words. Baseline is the clean input.

Accuracy and Flips

Table 1: Accuracy per config. *n* = questions; *n_answered* = questions that produced a final answer (not truncated at the token cap). *acc_answered* = correct / *n_answered*; *acc_strict* = correct / *n* (truncated counted as wrong); *ci_lo*/*ci_hi* = 95% bootstrap CI on *acc_strict*.

config	n	n_answered	acc_answered	acc_strict	ci_lo	ci_hi
clean	500	500	0.914	0.914	0.888	0.938
typo25_real10	500	497	0.8551	0.85	0.818	0.88
typo25_real40	500	500	0.87	0.87	0.838	0.898
typo25_real70	500	497	0.8672	0.862	0.832	0.892
typo50_real10	500	500	0.796	0.796	0.76	0.83
typo50_real40	500	500	0.79	0.79	0.754	0.824
typo50_real70	500	500	0.728	0.728	0.69	0.766
typo75_real10	500	488	0.7377	0.72	0.68	0.76
typo75_real40	500	495	0.6727	0.666	0.624	0.708
typo75_real70	500	480	0.6042	0.58	0.536	0.624

Table 2: Answered-only flips vs clean, paired by question over *n_both* (answered in both conditions). *r2w* / *w2r* = questions correct on clean but wrong on this config / wrong on clean but correct on this config; *r2w_ratio* = *r2w* / *n_both*, *w2r_ratio* = *w2r* / *n_both*; *net_flip_ratio* = *r2w_ratio* - *w2r_ratio* = (*r2w* - *w2r*) / *n_both*, the net rate at which this config worsens answers relative to clean.

config	n_both	r2w	w2r	r2w_ratio	w2r_ratio	net_flip_ratio	mcnemar_b	mcnemar_c	mcnemar_stat	mcnemar_p
typo25_real10	497	48	19	0.0966	0.0382	0.0584	48	19	11.701	0.0006245
typo25_real40	500	40	18	0.08	0.036	0.044	40	18	7.603	0.00582568
typo25_real70	497	43	19	0.0865	0.0382	0.0483	43	19	8.532	0.00348907
typo50_real10	500	78	19	0.156	0.038	0.118	78	19	34.68	0.0
typo50_real40	500	83	21	0.166	0.042	0.124	83	21	35.779	0.0
typo50_real70	500	108	15	0.216	0.03	0.186	108	15	68.813	0.0
typo75_real10	488	102	14	0.209	0.0287	0.1803	102	14	65.25	0.0
typo75_real40	495	136	17	0.2747	0.0343	0.2404	136	17	91.007	0.0
typo75_real70	480	164	14	0.3417	0.0292	0.3125	164	14	124.725	0.0

McNemar test (used in the flips table above).

- **What it is:** a paired significance test for before/after binary outcomes. It looks only at the questions that *changed* (the discordant pairs: *b* = right→wrong, *c* = wrong→right) and ignores those that stayed the same.

- **Null hypothesis:** typos break correct answers as often as they fix wrong ones ($b = c$, i.e. no net effect).
- **Statistic:** $(|b - c| - 1)^2 / (b + c)$ (column `mcnemar_stat`); `mcnemar_p` is its chi-square p -value with 1 degree of freedom.
- **When we reject it:** when `mcnemar_p` < 0.05 — the right→wrong vs wrong→right imbalance is too large to be chance.

Reasoning Length

Table 3: Absolute generated tokens per config (answered-only).

config	n_answered	mean_tok	median_tok	p90_tok	max_tok
clean	500	1582.3	1213.5	2951	9744
typo25_real10	497	1912.5	1255	4030	17255
typo25_real40	500	1814.4	1229.0	3910	12832
typo25_real70	497	1843.2	1259	3869	12031
typo50_real10	500	2023.5	1304.0	4561	17512
typo50_real40	500	2084.1	1299.5	4578	15205
typo50_real70	500	2094.3	1279.0	4781	14800
typo75_real10	488	2084.0	1407.5	3942	12803
typo75_real40	495	2593.5	1539	6311	13240
typo75_real70	480	2306.3	1326.5	5245	15252

Table 4: Mean generated tokens for correct vs wrong answers. `wrong_over_correct` = `tok_wrong` / `tok_correct` (mean tokens of wrong answers divided by mean tokens of correct answers).

config	tok_correct	tok_wrong	wrong_over_correct
clean	1406.1	3455.8	2.4578
typo25_real10	1511.2	4281.4	2.8331
typo25_real40	1566.9	3470.5	2.2149
typo25_real70	1537.0	3842.6	2.5
typo50_real10	1554.0	3855.7	2.4812
typo50_real40	1631.4	3786.8	2.3212
typo50_real70	1500.9	3682.5	2.4536
typo75_real10	1588.0	3479.0	2.1908
typo75_real40	1786.0	4253.2	2.3813
typo75_real70	1651.7	3305.5	2.0012

Self-doubt

The self-doubt markers are curated words grouped into two *banks*, each matched as a case-insensitive whole-word pattern in the reasoning text. Density is matches per 1000 reasoning words.

- **second_guess** — the proposal’s canonical self-correction words plus direct reversals: *wait, hmm, actually, reconsider, but wait, hold on, on second thought, let me recheck, double-check*.
- **uncertainty** — hedging / “not certain” words: *maybe, perhaps, possibly, might be, could be, not sure, I guess, presumably, or maybe / alternatively, unclear / confused*.

Example reasoning: “*Wait, maybe the total is 12... hmm, actually let me recheck*” contributes *wait*, *hmm*, *actually* (second_guess) and *maybe* (uncertainty). A separate *typo-noticing* bank (*typo*, *misspelled*, *doesn't make sense*, ...) is treated as repair behaviour, not self-doubt.

Table 5: Doubt-marker density per config: second_guess and uncertainty, per 1k words and per trace.

config	n_answered	second_guess_per_1k	second_guess_per_trace	uncertainty_per_1k	uncertainty_per_trace	total_per_1k	total_per_trace
clean	500	13.005	11.494	7.404	6.544	20.409	18.038
typo25_real10	497	14.742	15.954	12.321	13.334	27.063	29.288
typo25_real40	500	14.03	14.354	12.593	12.884	26.622	27.238
typo25_real70	497	14.525	15.294	13.151	13.847	27.677	29.141
typo50_real10	500	14.63	16.796	15.443	17.73	30.073	34.526
typo50_real40	500	14.698	17.484	15.861	18.868	30.559	36.352
typo50_real70	500	14.842	17.834	17.116	20.566	31.958	38.4
typo75_real10	488	14.425	17.057	18.708	22.121	33.133	39.178
typo75_real40	495	14.682	22.113	21.316	32.105	35.998	54.218
typo75_real70	480	14.062	18.867	21.924	29.415	35.986	48.281

Table 6: Per-marker density (markers per 1000 reasoning words): clean vs pooled-typo. discrimination = typo minus clean (larger = more typo-responsive). All bank markers, ranked by discrimination.

marker	category	clean_per_1k	typo_per_1k	discrimination
maybe	uncertainty	3.311	8.971	5.661
wait	second_guess	6.173	8.607	2.434
or maybe/alt	uncertainty	2.046	3.802	1.756
perhaps	uncertainty	0.86	1.988	1.128
unclear/confused	uncertainty	0.595	1.122	0.527
might be	uncertainty	0.124	0.312	0.187
not sure	uncertainty	0.054	0.24	0.186
not entirely	uncertainty	0.014	0.05	0.037
I assume	uncertainty	0.016	0.047	0.031
could be	uncertainty	0.154	0.175	0.021
presumably	uncertainty	0.002	0.013	0.01
possibly	uncertainty	0.0	0.005	0.005
but wait	second_guess	0.649	0.653	0.004
on second thought	second_guess	0.0	0.0	0.0
reconsider	second_guess	0.002	0.001	-0.001
hmm	second_guess	2.851	2.842	-0.01
actually	second_guess	0.405	0.332	-0.073
wait/actually no	second_guess	0.491	0.406	-0.085
I guess	uncertainty	0.229	0.102	-0.126
hold on	second_guess	0.758	0.601	-0.157
let me recheck	second_guess	0.763	0.521	-0.241
double-check	second_guess	0.912	0.558	-0.354

Table 7: Doubt density (markers per 1000 reasoning words) split by final outcome, per category (2g = second.guess, un = uncertainty, tot = both). For a group, density = total markers in that group / total reasoning words in that group x 1000. *_correct = over correct-answer traces; *_wrong = over wrong-answer traces; *_wrong_over_correct = _wrong / _correct.

config	2g_correct	2g_wrong	2g_wrong_over_correct	un_correct	un_wrong	un_wrong_over_correct	tot_correct	tot_wrong	tot_wrong_over_correct
clean	12.525	14.839		1.185	6.446	11.064	1.716	18.971	25.903
typo25_real10	13.673	16.847		1.232	9.622	17.637	1.833	23.294	34.484
typo25_real40	13.181	16.448		1.248	10.207	19.391	1.9	23.388	35.839
typo25_real70	13.794	16.296		1.181	9.914	20.991	2.117	23.709	37.287
typo50_real10	14.362	15.029		1.046	11.451	21.393	1.868	25.813	36.422
typo50_real40	14.439	15.087		1.045	12.677	20.642	1.628	27.115	35.729
typo50_real70	14.238	15.452		1.085	12.952	21.32	1.646	27.19	36.773
typo75_real10	13.945	15.006		1.076	14.607	23.665	1.62	28.552	38.671
typo75_real40	14.238	15.044		1.057	16.035	25.625	1.598	30.273	40.669
typo75_real70	13.344	14.577		1.092	16.607	25.732	1.549	29.951	40.309

LLM judge. The proposal specifies a second, non-lexical measure for both dimensions: a separate model with a fixed, strict prompt. The judge is a different model from the one under test, called at temperature 0 on a seeded random subsample of answered traces per config. It is shown the corrupted-word list (*original* $\rightarrow$ *as shown*), the question, and the reasoning trace (head+tail window). It returns two scalar ratings and evidence: *self_doubt_score* from 0–10, where 0 means no second-guessing and 10 means pervasive uncertainty or looping; and *repair_understanding_score* from 0–5, where 0 means the question’s meaning is fully understood and 5 means the model is fundamentally lost because of the corruption. The judge also returns a one-line justification and verbatim evidence quotes; quotes are checked against the reasoning trace. The judge is a subsample measure, while the lexical tables remain the full-data result.

Repair Behaviour

Table 8: How each corrupted word was handled, per config (categories defined below the table).

config	n_corrupt_words	silent_fix_n	flagged_n	misread_n	not_used_n	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
typo25_real10	3656	2678	435	106	437	0.7325	0.119	0.029	0.1195
typo25_real40	3673	2580	554	158	381	0.7024	0.1508	0.043	0.1037
typo25_real70	3650	2358	705	175	412	0.646	0.1932	0.0479	0.1129
typo50_real10	7202	4735	1344	340	783	0.6575	0.1866	0.0472	0.1087
typo50_real40	7179	4350	1677	375	777	0.6059	0.2336	0.0522	0.1082
typo50_real70	7173	4008	1814	475	876	0.5588	0.2529	0.0662	0.1221
typo75_real10	9833	5483	2565	674	1111	0.5576	0.2609	0.0685	0.113
typo75_real40	9945	4712	3268	827	1138	0.4738	0.3286	0.0832	0.1144
typo75_real70	9528	4403	2944	917	1264	0.4621	0.309	0.0962	0.1327

A *corrupted word* is a word that received a typo — it differs between the clean and typo question (e.g. *sum* $\rightarrow$ *sun*). We recover these by diffing the two questions, then place each into one bucket by which form appears in the reasoning:

- **silent_fix** — only the original word (*sum*) appears: a quiet recovery.
- **flagged** — both forms appear (*sum* and *sun*): noticed and fixed.
- **misread** — only the corrupted form (*sun*) appears: read as the wrong word.
- **not_used** — neither form appears in the reasoning.

Columns *_n are counts; *_frac are the share of n_corrupt_words.

Table 9: Word handling pooled by real-word ratio: the non-word vs real-word contrast.

real_ratio	n_corrupt_words	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
10	20691	0.6233	0.2099	0.0541	0.1127
40	20797	0.5598	0.2644	0.0654	0.1104
70	20351	0.5292	0.2684	0.077	0.1254

Table 10: Misread rate split by final correctness. A typo’d word is ‘misread’ when the reasoning uses only its corrupted form (e.g. sun instead of sum). Within each outcome group, misread rate = (typo’d words that were misread) / (all typo’d words examined), pooled over the group. misread_correct = over problems answered correctly; misread_wrong = over problems answered wrong; wrong_over_correct = misread_wrong / misread_correct.

config	misread_correct	misread_wrong	wrong_over_correct
typo25_real10	0.025	0.0499	1.993
typo25_real40	0.0392	0.0679	1.731
typo25_real70	0.0432	0.0774	1.79
typo50_real10	0.0389	0.0772	1.985
typo50_real40	0.0505	0.0583	1.155
typo50_real70	0.0607	0.0782	1.288
typo75_real10	0.0612	0.0869	1.422
typo75_real40	0.0676	0.1106	1.637
typo75_real70	0.0798	0.1193	1.495

Real-word Effect

This section isolates the effect of typo *kind* (real-word vs non-word), holding the question, corrupted positions, and rate fixed — something the vs-clean accuracy section cannot do. Three views:

- a **controlled paired** comparison of low-real vs high-real variants on the *same* questions, with a McNemar test (paired table);
- a **per-typo logistic regression** comparing the marginal harm of one real-word vs one non-word typo (logit table);
- whether wrong answers **flag the corruption** more or less as the real-word share rises (silent-failure table).

Read the signs, deltas and p -values in the tables below for this dataset’s result.

Table 11: Controlled paired comparison of two TYPO variants against EACH OTHER (baseline is the low-real variant, NOT clean – so this is not the vs-clean accuracy table). Within a fixed rate the same questions are corrupted in the same positions, only the typo KIND differs. Restricted to questions answered in both variants. $\text{acc_low_real} / \text{acc_high_real}$ = accuracy on the lower / higher real-word variant; $\text{delta_acc} = \text{acc_high_real} - \text{acc_low_real}$; r2w = questions correct in the low-real variant but wrong in the high-real one; w2r = the reverse; mcnemar_p = McNemar test on that paired change. Negative delta with $\text{r2w} \neq \text{w2r}$ means real-word typos are more harmful than the non-word typos they replaced.

rate	compare	n_both	acc_low_real	acc_high_real	delta_acc	r2w	w2r	mcnemar_p
25	real10_to_real70	494	0.8583	0.8684	0.0101	31	36	0.62506984
25	real10_to_real40	497	0.8551	0.8692	0.0141	28	35	0.4496918
25	real40_to_real70	497	0.8712	0.8672	-0.004	30	28	0.8955329
50	real10_to_real70	500	0.796	0.728	-0.068	76	42	0.00238236
50	real10_to_real40	500	0.796	0.79	-0.006	45	42	0.83021759
50	real40_to_real70	500	0.79	0.728	-0.062	70	39	0.00405978
75	real10_to_real70	468	0.7479	0.6068	-0.141	109	43	1.3e-07
75	real10_to_real40	483	0.7371	0.6791	-0.058	85	57	0.02346426
75	real40_to_real70	476	0.6849	0.6071	-0.0777	93	56	0.00318562

Table 12: Logistic regression $\text{correct} \sim \text{num_real} + \text{num_nonword}$ over all answered typo traces ($\text{num_real} / \text{num_nonword}$ = count of real-word / non-word typos in the problem). coef_* = change in log-odds of a correct answer per one added typo of that kind (more negative = more harmful); $\text{odds_*} = \exp(\text{coef})$; intercept is the model constant. n = number of traces fit.

n	coef_num_real	coef_num_nonword	odds_real	odds_nonword	intercept
4457	-0.07609	-0.03977	0.9267	0.961	2.09655

Table 13: Among WRONG answers only, does the model flag the corruption, by real ratio? $\text{explicit_notice_frac}$ = fraction of wrong-answer traces whose reasoning uses strong typo-flagging language (typo / misspelled / doesn't make sense / ...); $\text{silent_frac} = 1 - \text{explicit_notice_frac}$. The proposal predicts real-word failures are quieter ($\text{explicit_notice_frac}$ should fall as real rises).

real_ratio	n_wrong	explicit_notice_frac	silent_frac
10	302	0.7252	0.2748
40	332	0.738	0.262
70	392	0.7117	0.2883

Lexical Grid (overview)

Table 14: All four marker families (second_guess, uncertainty, typo_noticing, repair_words) across configs.

config	second_guess	uncertainty	typo_noticing	repair_words
clean	13.005	7.404	1.455	1.118
typo25_real10	14.742	12.321	2.945	2.253
typo25_real40	14.03	12.593	3.278	2.33
typo25_real70	14.525	13.151	3.394	2.484
typo50_real10	14.63	15.443	3.623	2.805
typo50_real40	14.698	15.861	3.749	2.865
typo50_real70	14.842	17.116	4.071	3.199
typo75_real10	14.425	18.708	4.261	3.253
typo75_real40	14.682	21.316	4.243	3.137
typo75_real70	14.062	21.924	4.287	3.373

Spellcheck Recovery